



Engineered design of self-cleaning and antifouling organosilicon/eugenol-based zinc acrylate resin composite coating by forming semi-interpenetrating polymer networks

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ABSTRACT

The practical application of low surface energy organosilicon in marine antifouling is significantly limited due to its poor ability to eliminate biofoulings. Herein, combining organosilicon with zinc acrylate resin containing eugenol group (EAM-Zn) by means of semi-interpenetrating polymer networks (SIPNs), an innovative self-cleaning and antifouling composite coating was developed (named E_xSi_y), which integrates low surface energy and self-polishing. After 35 days of immersion in seawater, the hydrophobic organosilicon of E₁Si₃ coating impedes the penetration of water molecules into the interior of the coating, resulting in the weight loss rate of EAM-Zn being reduced from 24.99 % to 1.92 %. The surface energy of the E_xSi_y coating after 7 days of immersion in seawater is decreased to 20 m J/m², achieving self-cleaning under the action of flowing water shear. Anti-fouling performance tests show that the E₁Si₁ coating increases the inhibition rate of the original silicone against *Escherichia coli* from 48.95 % to 77.5 %. The adhesion area of diatoms on the surface of the E_xSi_y coating is only 2 %, and the removal strength of pseudobarnacles is merely 0.2 MPa. Most significantly, the composite coating demonstrates a remarkable antifouling effect in the Yellow Sea waters of Qingdao. Overall, this research presents a preparation method for enhancing the antifouling performance of organosilicon antifouling coatings, promoting the development of multifunctional and environmentally-friendly marine antifouling coatings.

1. Introduction

Marine fouling organisms, including barnacles, algae, and mussels, cause a cascade of negative impacts [1] (such as slowing down the ship traveling speeds and accelerating the corrosion of metals) through their attachment to various marine facilities. Nowadays, among the numerous strategies [2,3] to prevent fouling organisms from attaching to the surfaces of marine facilities, applying marine antifouling coatings has emerged as the most effective approach to control marine biofouling [4]. However, antifouling coatings containing heavy metal ions can cause irreversible damages on organisms in the marine ecological environment, such as teratogenicity and growth inhibition [5,6]. Therefore, the research on environmentally friendly antifouling coatings has attracted extensive attention.

Organosilicon materials have attracted extensive attention in the

field of marine fouling release coatings [7,8] (FRCs), primarily attributed to their distinct features including low surface energy, low Young's modulus, and cost-effectiveness [9,10]. These properties can effectively reduce the adhesion between biofoulings and the interface, which can allow biofoulings to be easily detached under the action of shear forces [11–13] such as water flow and waves. However, in the complex marine environment, the hydrophobic nature of organosilicon renders it highly vulnerable to contamination by a complex mixture of slime-forming bacteria, proteins, and extracellular polymeric substances (EPS). These contaminants can adhere to the organosilicon surface, gradually reducing its surface energy-related antifouling efficiency [14]. Consequently, developing effective strategies to inhibit the growth of contaminants on the coating surface has emerged as a top-priority research focus.

In recent years, a variety of chemical substances, including metal

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ions [15,16], quaternary ammonium salts [17–19], zwitterionic materials [20–23] and natural antifouling agent [6,24,25], can effectively inhibit the growth of contaminants on the coating surface. Among them, metal ions and natural active substances have attracted extensive attention from researchers. Zinc ions, a type of metal ion, play a crucial role in disrupting bacteria's metabolic processes and cellular structures [26]. Yu [27] et al. prepared a polymer containing zinc ion by reacting rosin with zinc hydroxide, and this polymer exhibited an inhibition rate of over 95 % against bacteria and algae. Eugenol, as a natural antifouling agent, has been proven to have a remarkable inhibitory effect on a variety of bacteria [28] and excellent antifouling capabilities in the application scenarios of ships [6,29,30]. Based on these promising findings, Wang [21] et al. employed a simple one-step radical polymerization method to synthesize a self-polishing eugenol ester-based sily-acrylate polymer, which could effectively inhibit the growth of contaminants.

Therefore, to further enhance the ability of the silicone coating to inhibit contaminants, this study carefully designed an acrylic resin containing both eugenol groups and zinc ions (EAM-Zn), aiming to significantly improve the coating's performance. In this process, how to effectively combine these two different polymer systems has become a crucial issue that needs to be urgently addressed.

As is known to all, semi-interpenetrating polymer networks (SIPNs) which consist of polymer networks intertwined within linear polymers [31] represent a burgeoning area of research and development in materials science [32]. Consequently, this approach can effectively enhance various properties of polymer materials, including thermal stability [33], self-healing [34], chemical resistance [35] and mechanical behavior [36,37]. To strengthen the adhesion strength between the coating and surface, Qin [24] et al. harnessed the potential of SIPNs to craft a composite coating by integrating epoxy resin, polydimethylsiloxane (PDMS), and capsaicin, leading to a noticeable improvement in the antifouling performance of the coating in both static and dynamic marine environments. Therefore, SIPNs have the potential to efficiently combine EAM-Zn with silicone coatings, thereby forming new antifouling coatings.

In this study, we prepared a composite coating (E_xSi_y) with both EAM-Zn and organosilicon by two steps to further improve the inhibitory performance of the organosilicon coating against contaminants. First, the organosilicon is pre-crosslinked to form a stable network structure. Subsequently, it is compounded with the prepared linear eugenol-based zinc acrylate resin through SIPNs. Compared with the original organosilicon coating (E_0Si_1), the inhibition rate of the composite E_1Si_1 coating against *Escherichia coli* has significantly increased from 48.95 % to 77.50 %, indicating a substantial enhancement in antibacterial ability. The E_xSi_y coating also has significant advantages over the eugenol-based acrylate resin coating. The surface energy of the E_xSi_y coating remains at 20 mJ/m², the adhesion area of diatoms on the surface of this coating is only 2 %, and the removal strength of pseudobarnacle is 0.2 MPa. Notably, in actual marine environment tests, E_xSi_y coating also exhibited certain antifouling effects. In summary, our research providing a novel composite method for the development of multi-functional and environmentally friendly antifouling coatings. This is expected to drive technological innovation in the field of marine antifouling materials and provide strong support for the long-term protection of marine facilities.

2. Experimental

2.1. Materials

Eugenol ($C_{10}H_{12}O_2$, 99 %) and acrylic anhydride ($C_6H_6O_3$, 97 %) was purchased from Shanghai Adamas Reagent Co., Ltd. P-toluenesulfonic acid (TsOH, 99 %) was purchased from Shanghai Macklin Biochemical Technology Co., Ltd. 2,2'-Azobisisobutyronitrile (AIBN, AR) was purchased from Tianjin Damao Chemical Reagent Factory. Ethanol (C_2H_6O , AR)

was purchased from Sinopharm Chemical Reagent Co., Ltd. (Shanghai, China). Sodium bicarbonate ($NaHCO_3$, 99 %) was purchased from Aladdin Reagent (Shanghai) Co., Ltd.; Deionized water made by laboratory; Ethyl acrylate (EA), butyl acrylate (BA), acrylic acid (AA), methacrylic acid (MAA), xylene, n-butanol and butyl acetate were purchased from Datang Chemical Group Co. (Qingdao, China). Lauryl acrylate (LA) were provided from Guangzhou Swan Chemical Co., Ltd. (Guangzhou, China). Triisopropylsilyl acrylate (TIPSA) was purchased from Shin-Estu Chemical Co., Ltd. (Japan). RTV-66 organic silicone rubber were purchased from China Bluestar Chengrand Co., Ltd. (Chengdu, China). Methyltriaceoxysilane (METES, AR) was obtained from Shanghai Macklin Biochemical Co., Ltd.. Dibutyltin dilaurate (DBTDL, AR) was obtained from Sinopharm Chemical Reagent Co., Ltd. (Shanghai, China). Zinc oxide was purchased from Haishun New Material Co. (Zibo, China). Benzoic acid was purchased from Shanghai Yi'en Chemical Technology Co. (Shanghai, China). All chemicals are used directly without further purification.

2.2. Synthesis of eugenol acrylate monomer (EAM)

Refer to the steps described in Chinese Patent CN118930730A, Eugenol (8.21 g) and p-toluenesulfonic acid (0.22 g) were placed in a 100 mL three-way flask, stirred thoroughly, followed by the addition of acrylic anhydride (9.45 g) by esterification under the conditions of 40 °C for 24 h. After completion of the reaction, the resulting mixture was washed with a sodium bicarbonate aqueous solution (with a mass fraction of 5 %) and deionized water. The pH value of the solution was adjusted to neutral, then sequentially subjected to standing and filtration processes before freeze-drying to obtain yellow-brown solid Eugenol acrylate monomer (EAM).

2.3. Synthesis of acrylate prepolymer containing eugenol group (pre-EAM)

Pre-EAM was synthesized by free-radical copolymerization, the synthesis steps are shown in Fig. 1(b). Firstly, the monomer mixture (M1) was prepared by mixing 48.1 g BA, 22.5 g EA, 5.4 g AA, 6.5 g MAA, 5.1 g EAM, 9 g LA, and 4.3 g TIPSA. Secondly, 10 wt% of M1, solvent (xylene: n-butanol = 4:1, 150 wt% of M1), dodecanethiol (0.8 wt% of M1), and AIBN (0.15 wt% of M1) were added to a 500 ml three-necked flask. The reaction was carried out at 85 °C for 15 min. Afterwards, 90 wt % of M1 and AIBN (0.65 wt% of M1) were added to the three-necked flask at 85 °C in 9 equal portions at 15-minute intervals. After the intermittent addition, the reaction was carried out for 30 min and AIBN (0.20 wt% of M1) was added to the three-necked flask in two portions at 20 min intervals, and then reacted at 85 °C for 2 h to obtain prepolymers.

2.4. Synthesis of zinc acrylate resin containing eugenol group (EAM-Zn)

After prepolymer is cooled to 70 °C, 18.3 g benzoic acid, 12.3 g zinc oxide, and 33.3 g solvent (xylene: n-butanol: butyl acetate: water = 4.4: 1: 10.6: 1.5) were added to the prepolymer, and then raise the temperature to 90 °C for a 2-hour reaction to prepare the zinc / silicon acrylate resin containing eugenol group. The synthesis process is shown in Fig. 1(c).

2.5. Preparation of organosilicon / EAM-Zn composite coatings (E_xSi_y)

The E_xSi_y coating is prepared through the preparation method described in Fig. 1(d)(e) (where both x and y represent mass proportions, and the specific contents refer to the supporting documents). In the first step, at 25 °C, 10 g RTV-66, 150 μ L of DBTDL, and 1 g of METES [38] are stirred at 3000 rpm using a high-speed mixer for 10 min to achieve pre-crosslinking (mark as Si). In the second step, the prepared EAM-Zn is added according to the proportions in the table S1, and the mixture is stirred at 3000 rpm for another 10 min. Subsequently, the

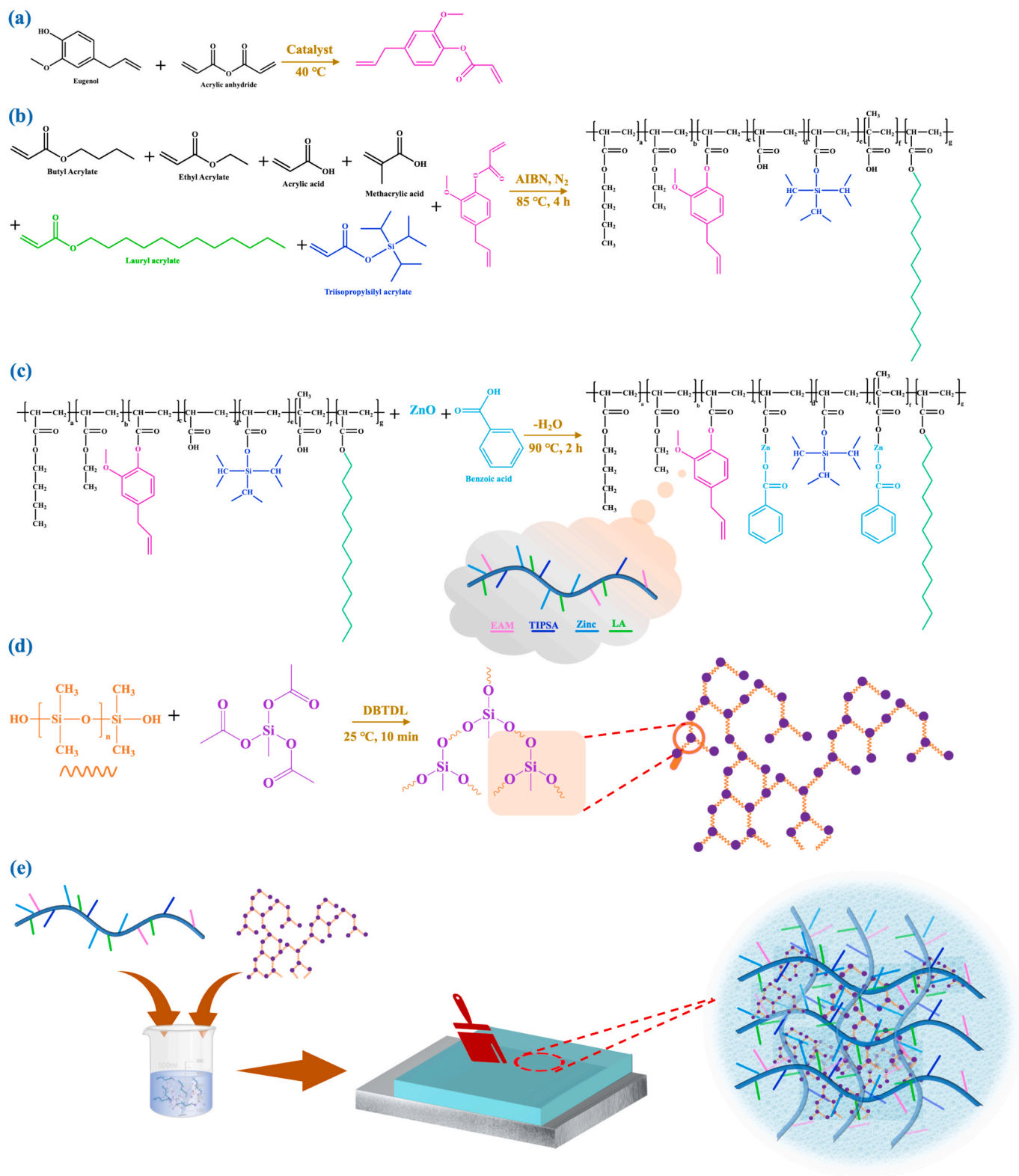


Fig. 1. Preparation of organosilicon / eugenol-based zinc acrylate resin composite coating (a) Synthesis route of the EAM. (b) Synthesis route of pre-EAM. (c) Synthesis of EAM-Zn polymer. (d) Pre-crosslinking of organosilicon. (e) Schematic diagram of the synthesis of E_xSi_y coatings.

mixture is brushed onto different substrates (ABS, PVC, glass fiber-reinforced epoxy, PET), and then cured for 24 h for subsequent experiments.

3. Characterization

3.1. Chemical structure analysis of EAM-Zn

3.1.1. Attenuated total reflection infrared (ATR-IR) Spectroscopy

Attenuated total reflection infrared (ATR-IR) spectra were recorded

on a German BRUKER TENSOR II transform infrared spectrometer. Powder/liquid/film samples were laid flat on a sample stage and the infrared absorption peaks at $4000 \sim 400 \text{ cm}^{-1}$ were measured. The test was performed against an air background.

3.1.2. Proton Nuclear Magnetic resonance spectroscopy (^1H NMR)

All ^1H NMR data were recorded on a Bruker AVANCE III 600 using deuterated chloroform (CDCl_3) as the solvent and tetramethylsilane (TMS) as internal standard.

3.1.3. X-ray photoelectron spectroscopy (XPS) measurement

The EAM-Zn polymer was applied to a polytetrafluoroethylene (PTFE) film, and after it dried, the film was cut into samples sized $5 \text{ mm} \times 5 \text{ mm}$. The samples were placed into the sample chamber of a Thermo Scientific K-Alpha XPS instrument. When the pressure in the sample chamber was less than $2.0 \times 10^{-7} \text{ mbar}$, the samples were introduced into the analysis chamber. The beam spot size was $400 \mu\text{m}$, with a working voltage of 12 kV and filament current of 6 mA . The element distribution on the coating surface was tested under the conditions of a full spectrum scan with an energy range of 150 eV and a step size of 1 eV , and a narrow spectrum scan with an energy range of 50 eV and a step size of 0.1 eV .

3.2. SEM and EDS elemental analysis of the coating cross-section

To study the morphology and composition of the coating cross-section, the coating cross-section was prepared using the low-temperature fracture method, and its morphology was observed by scanning electron microscopy. The specific operation steps are as follows: Place the sample to be tested in liquid nitrogen and freeze it for 10 min . Then, use an external force to break the frozen and hardened sample, and the coating cross-section to be tested can be obtained. Scanning electron microscopy (SEM) (HITACHI SU1000 FlexSEM1000II) was performed to investigate the morphologies of the coating cross-section which were sputtered with gold under vacuum prior to observation. Energy-dispersive X-ray spectroscopy (EDS) maps were recorded on the captured SEM images to analyze the elemental composition of the coating cross-section.

3.3. Hydrolysis properties

Prior to the immersion experiments, the $8 \text{ cm} \times 10 \text{ cm}$ ABS plates were cleaned with ethanol, then dried and weighed (W_0). The E_xSi_y resin was applied to the ABS boards, evaporated naturally for 48 h , and dried in an oven at 40°C for 48 h . The weight of each coated ABS board was recorded as W_t . The samples were immersed in real seawater, taken out every seven days, rinsed with deionized water, and dried in an oven at 40°C for 48 h , and the mass of the dried samples was recorded as W_i ($i = 1 \sim 5$). The percentage of the remaining mass of the coating after weight loss by immersion (W_s) was calculated by the formula.

$$W_s = \frac{W_i - W_0}{W_t - W_0} \times 100\%$$

3.4. Water absorption test

The water absorption [39] (W_A) property of the coating was evaluated by weighing. The composite resin was evenly applied onto a glass plate with dimensions of $2.5 \text{ cm} \times 7 \text{ cm}$ (the mass of the glass plate at this time was recorded as W_g). After 24 h of curing, a coating was formed on the surface of the glass plate (the mass of the glass plate with the coating was recorded as W_0). Subsequently, the glass plate with the coating was completely immersed in real seawater and taken out after 7 days quickly dried to remove the surface moisture, and weighed to obtain the W_t . The water absorption of each sample was calculated through the following equation.

$$W_A = \frac{W_t - W_0}{W_0 - W_g} \times 100\%$$

3.5. Tensile test

The tensile test referring to the GB/T 528–2009 was carried out on a SUNS-UTM6202 (SHENZHEN SUNS TECHNOLOGY STOCK CO.,LTD.) controller equipped with a 200 N load cell at a speed of 12 mm/min . The samples employed were Type 3 dumbbell-shaped specimens. At least five dumbbell-shaped specimens were tested for each coating sample to obtain an average value.

3.6. Testing of contact angle and surface free energy

The wettability of the coating surface is directly related to the wetting and spreading of the bio adhesive secreted by fouling organisms on the surface. It is an important parameter for evaluating the fouling release performance of the coating before and after immersion in real seawater for 7 days , and is usually characterized by the water contact angle and the surface free energy of the coating. An Optical Contact Angle Measuring Device (Dataphysics Contact Angle System OCA40, Germany) was utilized to investigate the surface wettability of the coatings under test. To ensure the accuracy of the results, five different positions were randomly selected on the same test sample, and the contact angle values were measured at each position. The average of these values was then calculated. The surface free energy of the coatings was calculated using the Owens-Wendt-Rabel-Kaelble (OWRK) geometric mean approach. Specifically, water and diiodomethane were employed to obtain the contact angle data of two types of droplets on the coating surface. By applying the OWRK formula, the required surface free energy values of the coatings were successfully determined.

3.7. Surface morphology and roughness

The coating was applied to $8 \text{ cm} \times 10 \text{ cm}$ ABS boards and allowed to dry naturally for 48 h , then oven dried at 40°C for 48 h . It was immersed in real seawater for 7 days . A laser confocal microscope (LEXT OLS4000) was used to observe the microscopic morphology and roughness of the coating surfaces and to evaluate the changes in morphology and roughness of the surfaces before and after static immersion in real seawater. R_a (arithmetic average roughness, which expresses the absolute value of the roughness over a reference length) was used to quantify the surface roughness of coatings.

3.8. Dynamic simulation experiment

The coating was applied to $250 \text{ mm} \times 100 \text{ mm} \times 3 \text{ mm}$ ABS boards with a controlled thickness of $400 \mu\text{m}$, allowed to dry naturally for 48 h , and dried at 40°C for 48 h . The sample boards were placed into a dynamic simulated paddling experimental setup (Fig. S1) containing natural seawater at a speed of 80 rpm , and the flatness of the coatings was tested under seawater washout conditions.

3.9. Pencil hardness test

Pencil hardness tests were performed on the coatings after 5 weeks of immersion in seawater, three times per sample, with reference to ASTM-D3363 test standard. Coating preparation reference hydrolysis performance test section.

3.10. Antifouling performance testing

3.10.1. Bacterial experiment

3.10.1.1. Antibacterial test. 5 mL of E_xSi_y was dropped onto an ABS disk

with a diameter of 3 cm. The disk was spin-coated using a spin coater (KW-4A, SETCAS LLC) at a speed of 3000 r/min for 3 s, then allowed to dry naturally for 48 h and finally dried in a 40 °C oven for 48 h.

To systematically evaluate the broad-spectrum antibacterial performance of E_xSi_y coatings, we employed a standardized plate count method for quantitative assessment of the antimicrobial efficacy as a function of organosilicon content. The experimental design incorporated two representative bacterial strains selected based on fundamental Gram-staining [40] characteristics: *Staphylococcus aureus* (*S. aureus*, Gram-positive) and *Escherichia coli* (*E. coli*, Gram-negative). These model organisms were chosen to represent distinct bacterial membrane architectures—*S. aureus* possesses a thick peptidoglycan layer reinforced with teichoic acids, while *E. coli* exhibits a more complex cell envelope structure comprising an outer membrane rich in lipopolysaccharides overlaying a thinner peptidoglycan layer. This comparative approach enabled comprehensive analysis of the coating's differential inhibitory effects against both Gram classifications, thereby facilitating robust evaluation of its broad-spectrum antibacterial properties. The findings provide valuable mechanistic insights for the rational design of marine antifouling materials with enhanced antimicrobial performance.

The experimental method is as follows: initially, a single colony was cultured in nutrient broth at 37 °C for 24 h to obtain an initial bacterial suspension with a concentration of approximately 10^9 CFU/mL; then, at 37 °C, each sample coating was immersed in 5 mL of the initial bacterial suspension for 24 h, allowing the bacteria to come into contact with the coating and potentially be affected by its antibacterial components; after immersion, the bacterial suspension containing the coated samples was diluted to 10^{-5} using the tenfold serial dilution method to reduce the bacterial concentration and facilitate the counting of colonies on solid media; finally, one hundred microliters of the diluted bacterial suspension were evenly spread onto solid culture media, which were then incubated at 37 °C for 24 h. The colony formation was observed and recorded to compare the antibacterial effects of different sample coatings. To quantify the antibacterial performance of the coating, we calculated the antibacterial rate (A) using the following formula:

$$A = \frac{N_0 - N_i}{N_0} \times 100\%$$

Where N_0 and N_i represent the colony counts of the blank control sample and the experimental sample, respectively. A higher antibacterial rate (A) indicates better antibacterial properties of the coating. Throughout the experiment, all materials used were sterilized in an autoclave to ensure sterile conditions. Additionally, all operations were performed in a sterile laminar flow cabinet to prevent external contamination from affecting the experimental results.

3.10.1.2. Coating antibacterial adhesion experiment. The antibacterial adhesion rate is a quantitative indicator used to measure the ability of a coating or material to resist bacterial adhesion. This test selected the *Escherichia coli* (Gram-negative bacterium) as the test strain. The E_xSi_y coating applied by spin-coating was immersed in 5 mL of a bacterial solution with a concentration of 10^9 CFU/mL at 37 °C for 24 h. After taking out the coating that had been immersed in the bacterial solution, a cotton swab was used to absorb the bacterial solution that had not adhered to the surface. Then, 8 mL of normal saline was used to rinse the bacterial solution on the coating surface into a six-well plate. Subsequently, the diluted multiple method was employed to dilute the rinsed bacterial solution to 10^{-3} . 100 μ L of the diluted bacterial suspension was evenly coated on a solid medium, and then cultured at 37 °C for 18 h. The growth of the bacterial colonies was observed and recorded. In order to evaluate the adhesion of bacteria to the coating, the antibacterial adhesion rate of the coating was calculated using the following formula.

$$S = \frac{R_0 - R_i}{R_0}$$

Where R_0 and R_i represent the colony counts of the blank control sample and the experimental sample, respectively. A higher antibacterial adhesion rate indicates that it is more difficult for bacteria to adhere to the surface of the coating. Throughout the experiment, all materials used were sterilized in an autoclave to ensure sterile conditions. Additionally, all operations were performed in a sterile laminar flow cabinet to prevent external contamination from affecting the experimental results.

3.10.2. Diatom attachment experiment

Diatoms, being the primary fouling organisms in the early stage of fouling initiation, enable the investigation of a coating's antifouling performance through observing their attachment onto the coating surface. In this thesis, *N. closterium* was employed to study the antifouling properties of the coating materials under test [8]. The detailed experimental procedures are as follows: The cultivation of diatoms was carried out in accordance with the diatom active growth experiment protocol. After that, the coating samples, which had been immersed in the diatom culture solution for three days, were retrieved. The samples were gently rinsed with artificial seawater to eliminate the diatom culture medium on the coating surface and the diatoms that were not firmly adhered. The rinsed samples were then examined under a fluorescence microscope (SOPTOP ICX41, NINGBO SUNNY INSTRUMENTS CO.,LTD). With a 40-fold objective lens, ten positions were randomly selected on the sample surface for observation and photography. The attachment state of diatoms on the coating surface was recorded, and the acquired images were processed using ImageJ software.

3.10.3. Pseudobarnacle adhesion test.

Pseudobarnacle adhesion measurement was performed according to ASTM Standard D 4541. An aluminous cylindrical stud with a diameter of 20 mm and a height of 10 mm was glued with epoxy adhesive to the coating surface on the glass fiber reinforced epoxy resin panel. The adhesive was cured under ambient conditions for 7 days. The stud was moved in a vertical direction to the coated panel at a certain speed of 0.2 Mpa/s, and the shear force required to dislodge the stud was measured by using a force gauge (PosiTest AT-A Automatic). All tests were conducted by the same operator to minimize the standard error.

3.10.4. Marine field test

The anti-fouling performance of the coating was tested according to the Chinese standard GB/T 5370-2007 "Test Method for Anti-fouling Paint Sample by Shallow Sea Immersion". The offshore PVC wallboard frames designed by the laboratory were divided into 14 pieces, and each piece was coated with the polymer coating with an area of 140 mm x 160 mm and a thickness of 150–200 μ m, and left to dry in a ventilated place. The antifouling performance of the composite coating was evaluated through marine field tests conducted in Qingdao Middle Harbor (36.1°N, 120.3°E) from December 19, 2024 to January 19, 2025. During the testing period, the coated frames were immersed in natural seawater at depths ranging from 0.5 to 2.0 m, where the water temperature fluctuated between 5 ~ 10 °C and the pH remained stable within 7.8 ~ 8.4. In order to ensure the reliability of the results of the experiments, a blank PVC board was chosen as the control group, and samples were taken at regular intervals within the specified time range, so that the growth of fouling organisms could be recorded. Evaluate the antifouling ability of the resin coating. In addition, all the samples should be put back into the seawater immediately after taking photos to prevent the organisms from dying in the air for too long.

4. Results and discussion

4.1. Structural characterization

To gain in-depth insights into the structural characteristics of EAM, ^1H NMR and ATR-IR were employed for its characterization. In the

spectrum shown in Fig. 2(a), a resonance peak corresponding to the double bond (i, k) is clearly presented at a chemical shift of 6.5 ppm. Focusing on the ATR-IR spectrum in Fig. 2(b), through meticulous comparative analysis, it is found that the characteristic absorption peak of the phenolic hydroxyl group originally attributed to eugenol at 3450 cm^{-1} completely disappears in the spectrum of the EAM compound. At the same time, a peak consistent with the characteristic absorption of the C=O group in acrylic anhydride appears at 1750 cm^{-1} , indicating that the C=O structure of acrylic anhydride is successfully introduced into the EAM molecular system. In addition, an obvious -C-O-C- absorption peak [29] is detected at approximately 1291 cm^{-1} , which further confirms the existence of the corresponding ester groups in the EAM. Therefore, these evidences strongly implies that the phenolic hydroxyl group of eugenol participates in the formation of new chemical bonds during the synthesis process of EAM.

A prepolymer (pre-EAM) with -COOH in the side chain was successfully obtained by copolymerizing EAM with acrylates. Subsequently, the successful preparation of the EAM-Zn resin was achieved by adding ZnO and benzoic acid to the pre-EAM followed by a dehydration reaction. To deeply explore the compositional characteristics of EAM-Zn resin, a systematic characterization analysis was carried out. Firstly, pre-EAM was detected by means of ^1H NMR. The observation of characteristic signal peaks corresponding to the substituted hydrogens on the benzene ring within the chemical shift range of 6.01 ~ 6.82 ppm serves as compelling evidence that EAM has successfully engaged in the free-radical copolymerization process.

Secondly, XPS technology was employed to conduct a meticulous analysis of the elemental composition and chemical bonds of EAM-Zn resin. Through accurate determination of the elemental content, it was determined that the resin contained 6.79 % Si and 0.63 % Zn (Fig. 2 (d)). Focus on the C 1 s XPS spectrum shown in Fig. 2(e), four peaks can be spotted in the C 1 s XPS spectrum [29]. The peak located at 284.78 eV is assigned to the C-C bonds in the main chain and C=C bonds in EAM. The peaks at 286.58 eV and 288.78 eV is respectively known as the C-O bond and C=O bond from acrylates. The peak at 291.98 eV is deemed to the benzene -ring bond in the resin [30]. The O1s spectrum (Fig. 2 (f)) shows two peaks at 531.98 and 533.48 eV, which can individually be attributed to C-O and C=O structure. The Si2p spectrum (Fig. 2 (g)) shows one peak at 102.08 eV which can be matched to Si-O structure [41,42] from TIPSA. As presented in Fig. 2(h), the two peaks of the Zn2p

spectrum located at 1022.38 eV and 1045.38 eV are ascribable to the binding energy of Zn 2p_{1/2} and Zn 2p_{3/2} which matches with Zn (COO)₂ and ZnO [43]. In conclusion, it has been strongly confirmed at the molecular and elemental levels that the EAM-Zn resin has been successfully prepared.

4.2. SEM and EDS elemental analysis of the coating cross-section

Conducting a detailed observation of the cross-sectional morphology of the prepared coating can provide a highly intuitive reflection of crucial information such as the compatibility among various components within the resin system and the mechanical properties of the material itself. In Fig. 3a, after the solvent in the eugenol-based zinc acrylate resin coating has completely volatilized, its cross-section exhibits an obviously uneven morphology, and there are also a certain number of pores. These pore structures provide extremely favorable conditions for the subsequent compounding process.

When the silicone rubber is compounded with the EAM-Zn resin, it can be clearly observed from Fig. 3b that the silicone rubber effectively fills the pores of the EAM-Zn resin coating. However, due to the certain limitations in the compatibility between the two materials, during the curing process, the EAM-Zn polymer and the silicone elastomer entangle with each other, ultimately resulting in an irregular and uneven surface of the coating (Fig. 3c). As the mass fraction of EAM-Zn resin in the entire system gradually decreases, the cross-section of the E₁Si₅ coating tends to be flat (Fig. 3d). However, when the content of EAM-Zn decreases to 16.6 % (E₁Si₅), the EAM-Zn resin in the coating is completely encapsulated by the elastomer inside, and thus a unique protrusion structure is formed (Fig. 3e). In summary, under appropriate proportion conditions, silicone elastomer can be well compounded with the zinc acrylate resin system which not only effectively enhances the hydrophobicity of the coating but also significantly improves its flexibility.

4.3. Hydrolysis properties

In the marine environment, the eugenol-based zinc acrylate coating (E₁Si₀) undergoes a certain degree of hydrolysis reaction. Within just 35 days, its mass loss reaches as high as 25 % (Fig. 4a, b). This is primarily attributed to the high humidity and high salinity characteristics of the marine environment. Under the continuous attack of water molecules,

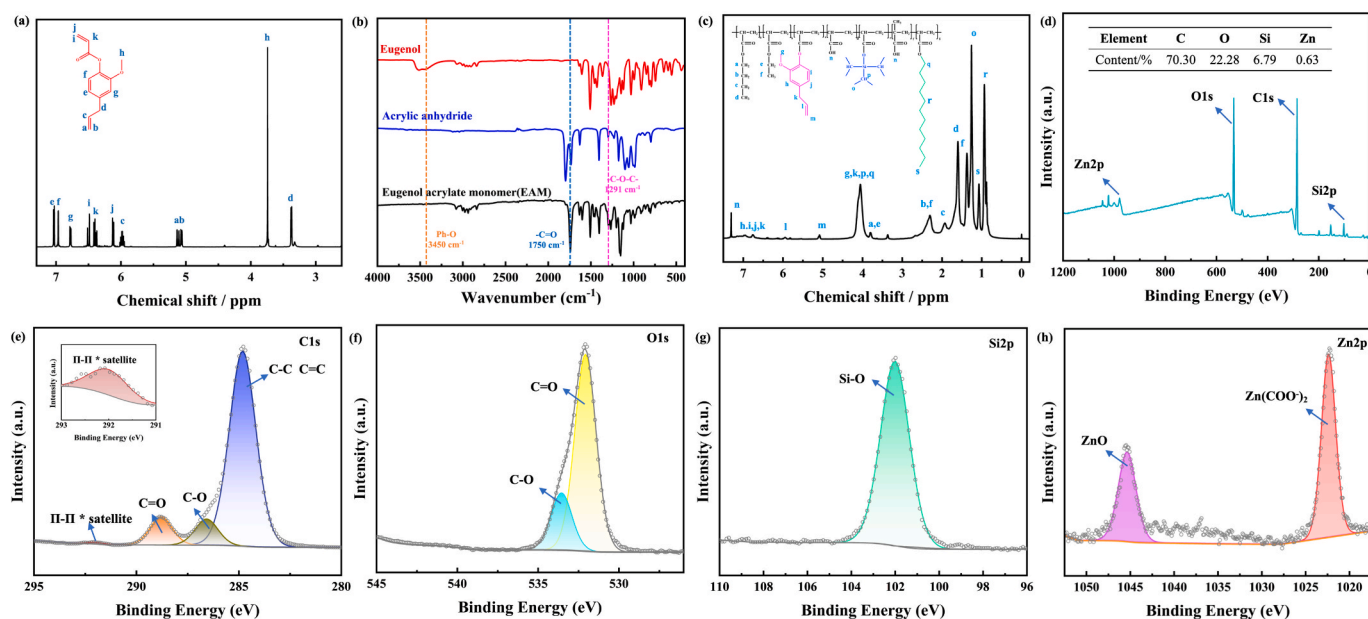


Fig. 2. (a) ^1H NMR spectra of EAM. (b) ATR-IR spectra for EAM, eugenol and acrylic anhydride. (c) ^1H NMR spectra of pre-EAM. (d) XPS spectrum of EAM-Zn. (e-h) High resolution C1s(e), O1s(f), Si2p(g) and Zn2p(h) spectra of EAM-Zn.

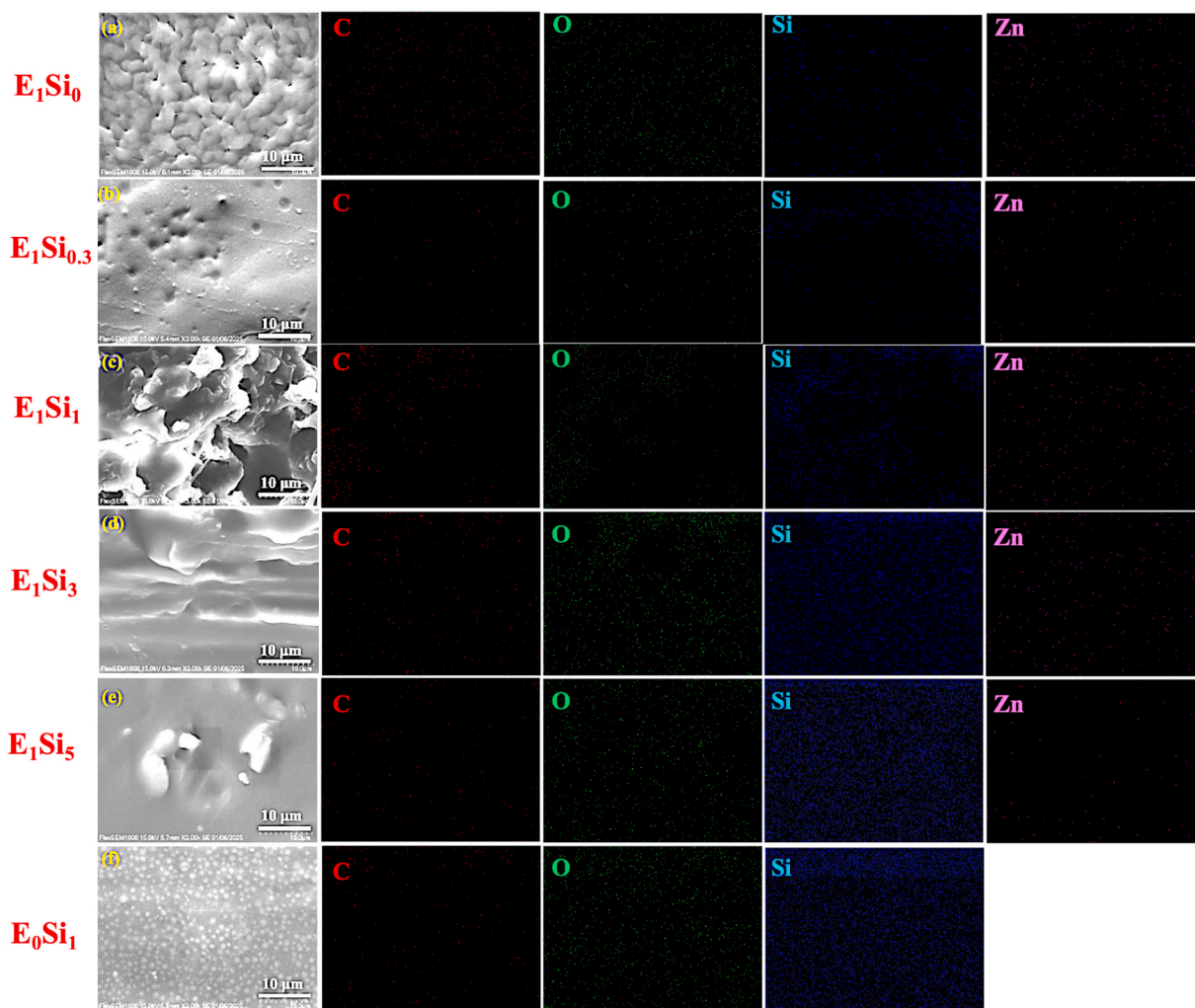


Fig. 3. SEM and EDX images of (a)E₁Si₀ (b)E₁Si_{0.3} (c)E₁Si₁ (d)E₁Si₃ (e)E₁Si₅ (f)E₀Si₁. The Zn signal in pink is associated with EAM-Zn.

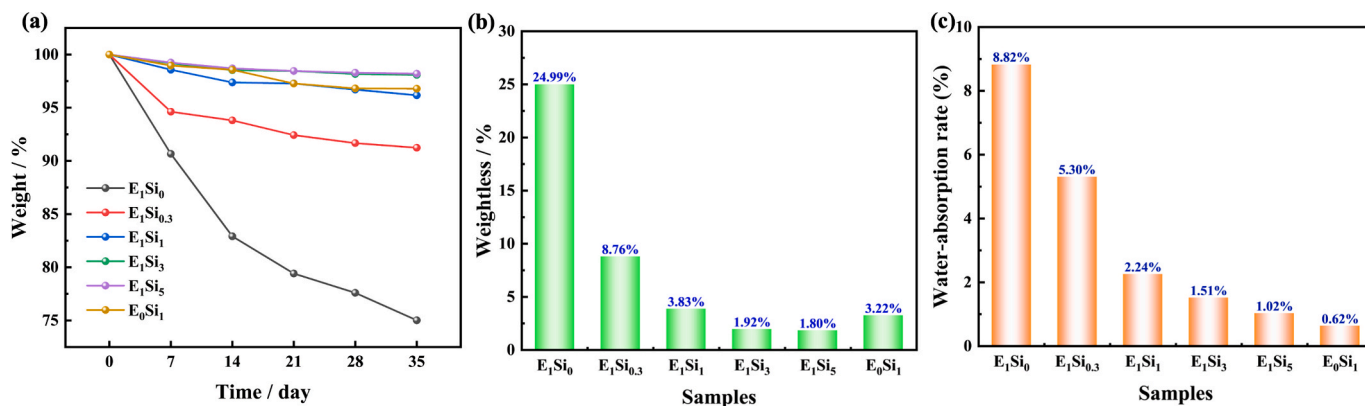


Fig. 4. (a) Mass change of ExSiY coatings over 35 days (b) Mass loss rate of ExSiY coatings over 35 days (c) Water absorption rate of ExSiY coatings within 7 days.

the chemical bonds of eugenol, acrylate, and zinc ions in the coating break, leading to the dissolution or detachment of some substances and thus causing a significant loss of mass. However, when organosilicon (Si) with their main chain composed of Si-O bonds which have high bond energy and excellent hydrophobicity is introduced into this coating system, thereby blocking the contact between external water molecules and the internal chemical bonds of the coating, and reducing the water absorption rate of the coating (Fig. 4c).

Consequently, as the organosilicon content gradually increases, the mass loss rate of the coating shows a distinct downward trend (Fig. 4b). In particular, when the organosilicon content is increased to 50 %, the hydrolysis rate drops sharply to 2 %. In addition, the introduction of organic silicon can also affect the surface morphology of ExSiY coatings, thereby influencing the contact between the coating surface and water molecules. Surface characterization (Fig. 6) reveals that coatings with lower surface roughness (E₁Si₃, Ra = 0.28 μm; E₁Si₅, Ra = 0.59 μm)

demonstrate superior water resistance, attributable to their smooth, dense morphology (Fig. 6 (j), (k)) and reduced porosity that effectively limit water penetration pathways.

For coatings with higher surface roughness, the competitive effects between surface topography and chemical hydrophilicity/hydrophobicity become particularly important. For instance, although E_3Si_1 ($R_a = 5.82 \mu m$) and E_1Si_1 ($R_a = 2.03 \mu m$) possess micro-nano scale rough surfaces that could theoretically enhance hydrophobicity through the “lotus effect”, the presence of hydrophilic groups (such as eugenol and ester groups) in their composition promotes water adsorption and internal penetration, resulting in higher water absorption rates compared

to the more chemically inert E_0Si_1 ($R_a = 3.83 \mu m$).and further achieve the controlled release of substances in the coating that have an inhibitory effect on marine organisms, providing an effective strategy for the design of marine antifouling coatings.

These findings have important implications for marine antifouling applications. The dual control mechanism—combining the physical barrier effect of the Si-O-Si network with tailored surface morphology—not only improves water resistance but also enables controlled release of marine biofouling inhibitors. This synergistic approach provides a robust strategy for designing advanced antifouling coatings with optimized performance characteristics.

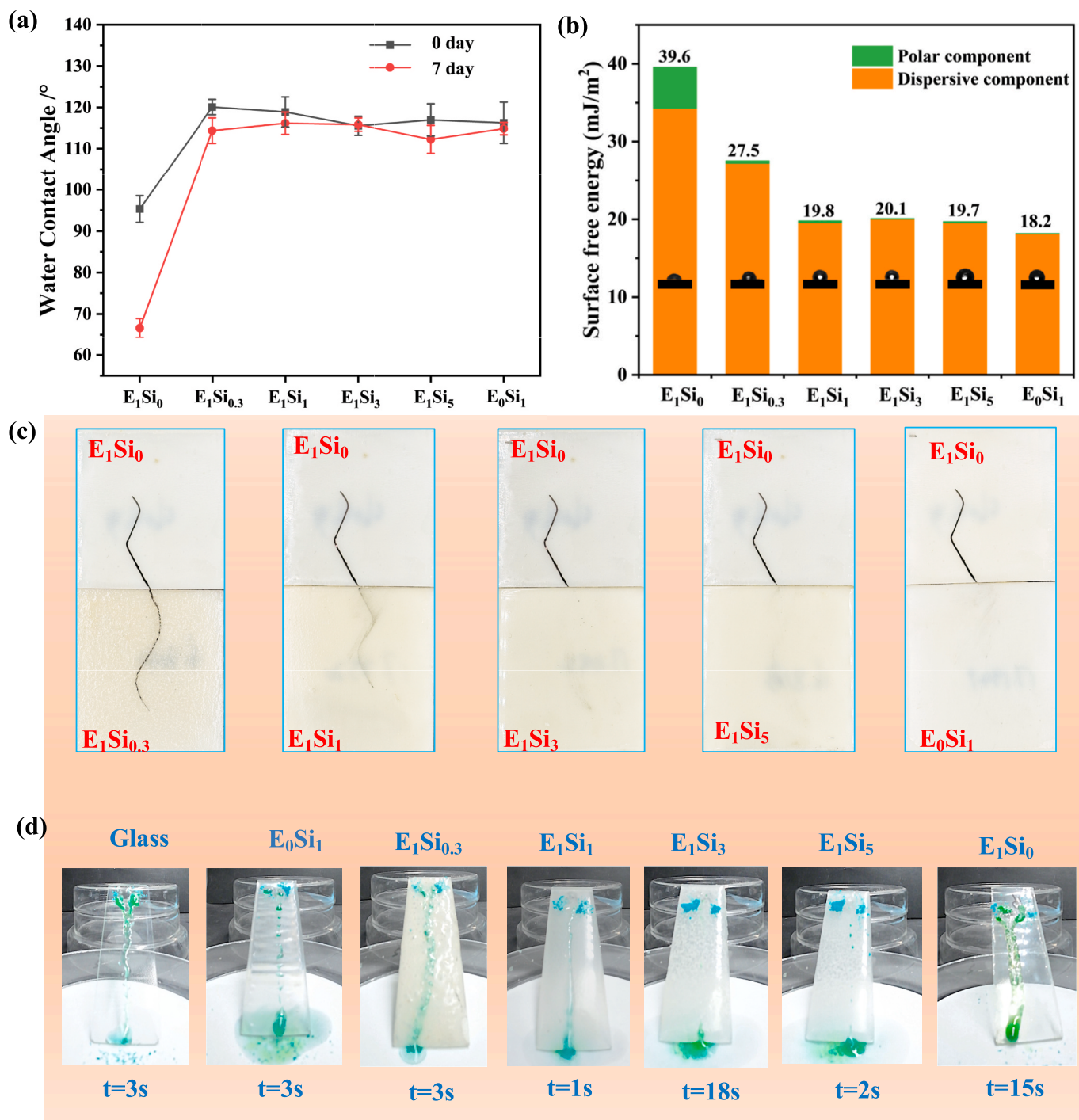


Fig. 5. (a) Contact angle changes of E_xSi_y coatings before and after 7-day immersion in seawater. (b) Surface energy of E_xSi_y coatings after 7-day immersion in seawater. (c) Anti-graffiti performance test of E_xSi_y coatings. (d) Self-cleaning performance test of E_xSi_y coatings.

4.4. Testing of contact angle and surface free energy

The surface wetting property of a coating directly affects its surface energy level and the resistance to various types of fouling. Therefore, detecting the surface wettability and surface energy of a coating is a crucial step in evaluating the antifouling ability of the coating. As shown in Fig. 5, when the EAM-Zn (E_1Si_0) coating is immersed in a seawater environment for 7 days, its contact angle drops to 66.7° and the surface energy reaches 39.6 mJ/m^2 . As a result, the coating shows hydrophilic characteristics, making it hard to achieve effective self-cleaning and anti-graffiti functions [44].

In marked contrast, the contact angle of the organosilicone coating (E_0Si_1) remains steadfastly at 115° before and after being immersed in seawater and the surface energy of coating after 7 days of immersion is 18.2 mJ/m^2 (Fig. 5 (a),(b)). According to the Bayer curve [45], coatings with a surface free energy lower than 25 mJ/m^2 can achieve self-cleaning under the action of flowing water shear. Therefore, the organosilicone coating (E_0Si_1) has excellent self-cleaning performance. Fig. 5d exhibits coatings' remarkable self-cleaning properties, and the result shows a series of E_xSi_y coatings can achieve the cleaning of contaminants through the action of external shear forces. In summary, the enhancements in hydrophobic performance lay a solid foundation for the composite coatings to achieve the release of fouling organisms.

Furthermore, the hydrophobic properties of the composite coating originate not only from the low surface energy conferred by the polysiloxane structure of the organosilicone component but are also significantly influenced by surface roughness. According to the Wenzel model [46], when the surface roughness factor exceeds 1 and the contact angle θ surpasses 90° , increased surface roughness further amplifies the contact angle, thereby enhancing hydrophobicity. This theoretical framework effectively explains the distinctive behavior observed in the $E_1Si_{0.3}$ coating: despite its reduced organosilicone content compared to the E_1Si_1 coating, the contact angle remains consistently around 115° due to a substantial increase in surface roughness from $2.03 \mu\text{m}$ to $5.82 \mu\text{m}$. This augmented surface roughness not only improves the coating's hydrophobic performance but also imparts superior self-cleaning capability, enabling effective repulsion of contaminants [47]. These findings provide crucial insights into the performance mechanisms of composite coatings while demonstrating excellent agreement with established theoretical models.

4.5. Surface morphology and roughness

The surface morphology of the coating is one of the key indicators for measuring its performance and plays a decisive role in the stability of the coating. As depicted in Fig. 6, a laser confocal microscope was used to

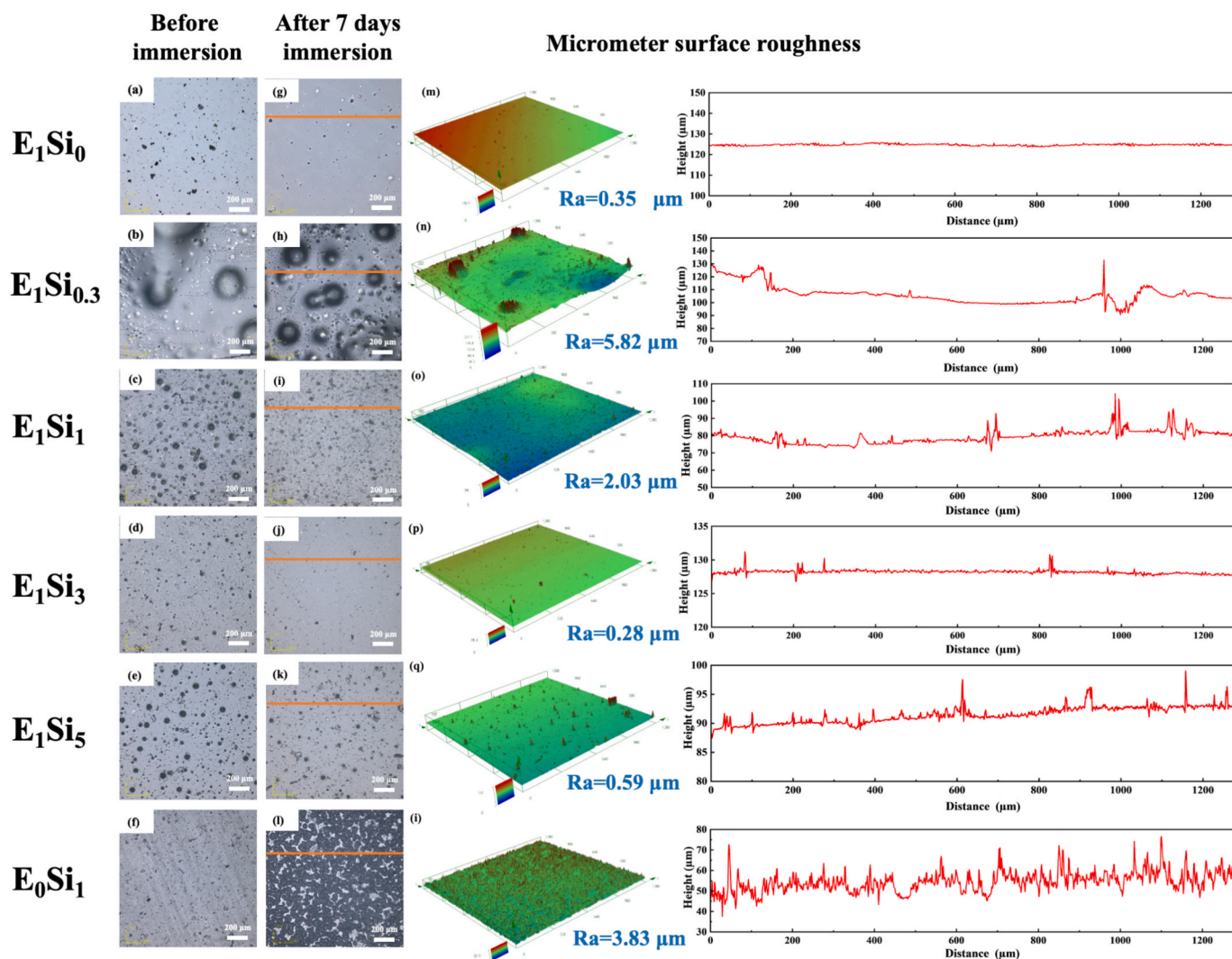


Fig. 6. Representative laser confocal images of E_xSi_y coatings (a-f) before immersion and (g-l) after 7 days immersion in real seawater; and (m-i) representative 3D images and roughness (Ra) of E_xSi_y coatings after 7 days immersion in real seawater.

meticulously characterize and analyze the changes in the micron-scale morphology and roughness of the coating before and after being immersed in real seawater for seven days, aiming to deeply explore the performance evolution laws of the coating in the actual marine environment.

Integrating the test results of the coating film-forming property presented in Fig. 7c, when the organosilicon addition amount reaches 25 % ($E_1Si_{0.3}$), it becomes evident that the combination of organosilicon and EAM-Zn has a significant impact. This impact leads to a marked deterioration in the film-forming quality, ultimately resulting in an uneven coating surface with a roughness R_a of $5.82\text{ }\mu\text{m}$, which clearly impairs the coating's film-forming performance. As the organosilicon content increases to 75 %, the microscopic morphology of the E_1Si_3 coating surface appears smooth and flat, with the roughness reaching a minimum value ($R_a = 0.28\text{ }\mu\text{m}$). These collective observations definitively demonstrate that organosilicon content serves as a critical determinant governing both the microstructural architecture and macroscopic performance characteristics of the coating surface, with this intricate interplay ultimately manifesting as dynamic evolution in surface roughness.

The alteration of coating surface roughness profoundly influences its interaction mechanisms with aqueous environments. On one hand, precise modulation of surface roughness can optimize the contact mode between the coating and water, thereby significantly reducing the fluid dynamic drag exerted on the coating in complex marine flow fields. This is of great significance for improving the navigation efficiency of marine equipment such as ships. On the other hand, changes in roughness also induce notable variations in the surface free energy state of the coating, promoting the formation of smoother and more hydrophobic characteristics (as shown in Fig. 5). Notably, the observations from Fig. 8(c) (c') reveal that at surface roughness values of $0.28\text{ }\mu\text{m}$ (E_1Si_3) and $0.59\text{ }\mu\text{m}$ (E_1Si_5), the diatom coverage areas are reduced to only 1.98 % and 2.34 %, respectively—significantly lower than those on coatings with other roughness levels. This finding strongly confirms the intrinsic correlation between coating surface roughness and antifouling performance. Research indicates that appropriately reducing surface roughness can

create a physical microenvironment unfavorable for marine biofouling, effectively suppressing the adhesion probability and coverage area of marine organisms such as diatoms on the coating surface.

These research outcomes provide crucial theoretical and experimental foundations for the development of high-performance marine antifouling coatings, which will significantly advance the innovation and progress of marine antifouling technologies.

4.6. Testing of mechanical properties and transparency

As a crucial component of mechanical properties, tensile properties play a vital role in the flexibility of the coating [48]. Coatings with excellent tensile properties can largely avoid cracking problems.

As shown in Fig. 7b, upon the introduction of organosilicon into the EAM-Zn system, the E_xSi_y coating's fracture strength experiences a drastic decline while still maintaining a certain elongation degree. Although the E_1Si_1 coating with an organosilicon content of 25 % has a relatively low fracture strength, it exhibits extremely excellent elongation, with a value exceeding 400 %. This fully demonstrates that when the proportion of organosilicon components is relatively small, the coating has good flexibility and can effectively prevent the occurrence of cracking. In sharp contrast, the E_1Si_1 coating has an organosilicon content as high as 50 %, causing its fracture strength to drop to the lowest level. As a result, under the continuous action of external forces like seawater scouring, the E_1Si_1 coating is extremely prone to damage (Fig. 7 d), significantly shortening its service life.

The hardness of the coating plays a crucial role in determining its wear resistance, impact resistance, lubricity, and other properties. As can be seen from Fig. 7d, after adding silicone to the E_xSi_y coating, the hardness drops sharply which indicates that the surface state of the composite coating is closer to that of the silicone coating. Although the hardness is reduced, under the scouring of seawater, it can still maintain a flat surface morphology without obvious scratches. This implies that it exhibits remarkable flexibility, which offers a novel perspective for the application of antifouling coatings on irregular surfaces.

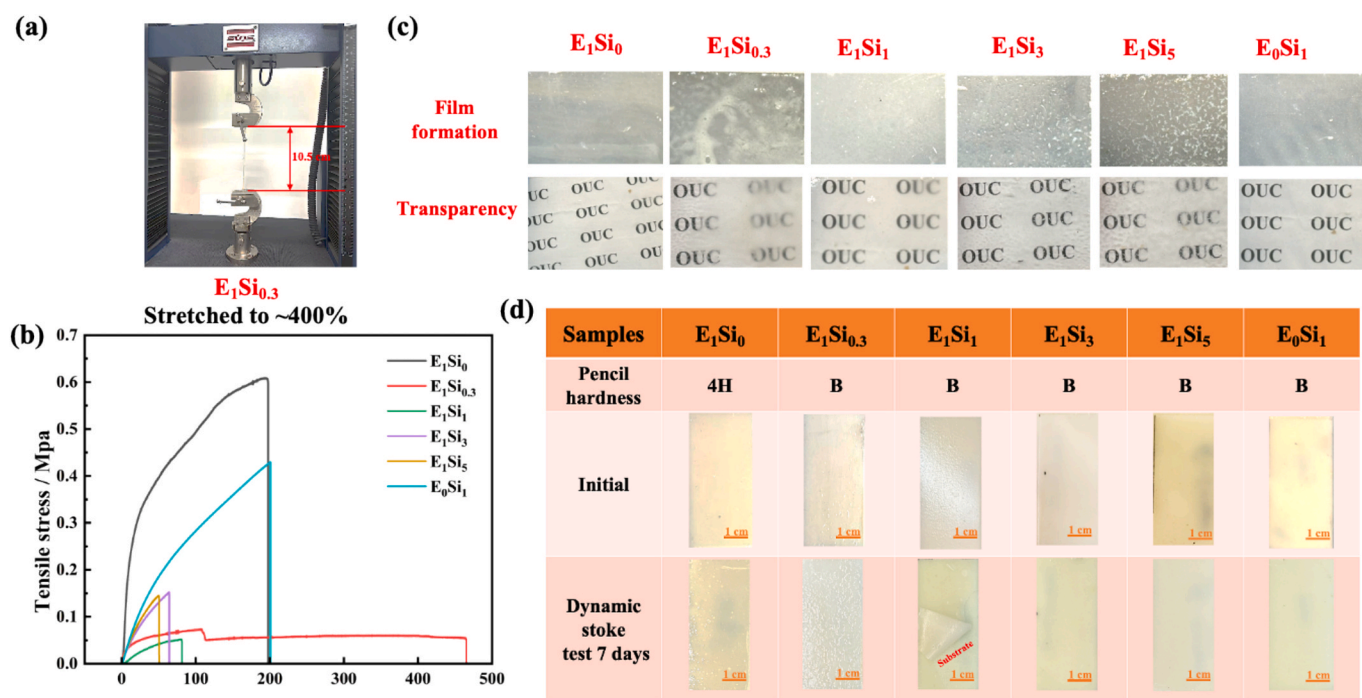


Fig. 7. (a) Photos of the stretchability of the $E_1Si_{0.3}$. (b) Stress-strain curves of the E_xSi_y coating. (c) Transparency of the E_xSi_y coating. (d) The flatness of the E_xSi_y coating before and after water erosion under dynamic conditions.

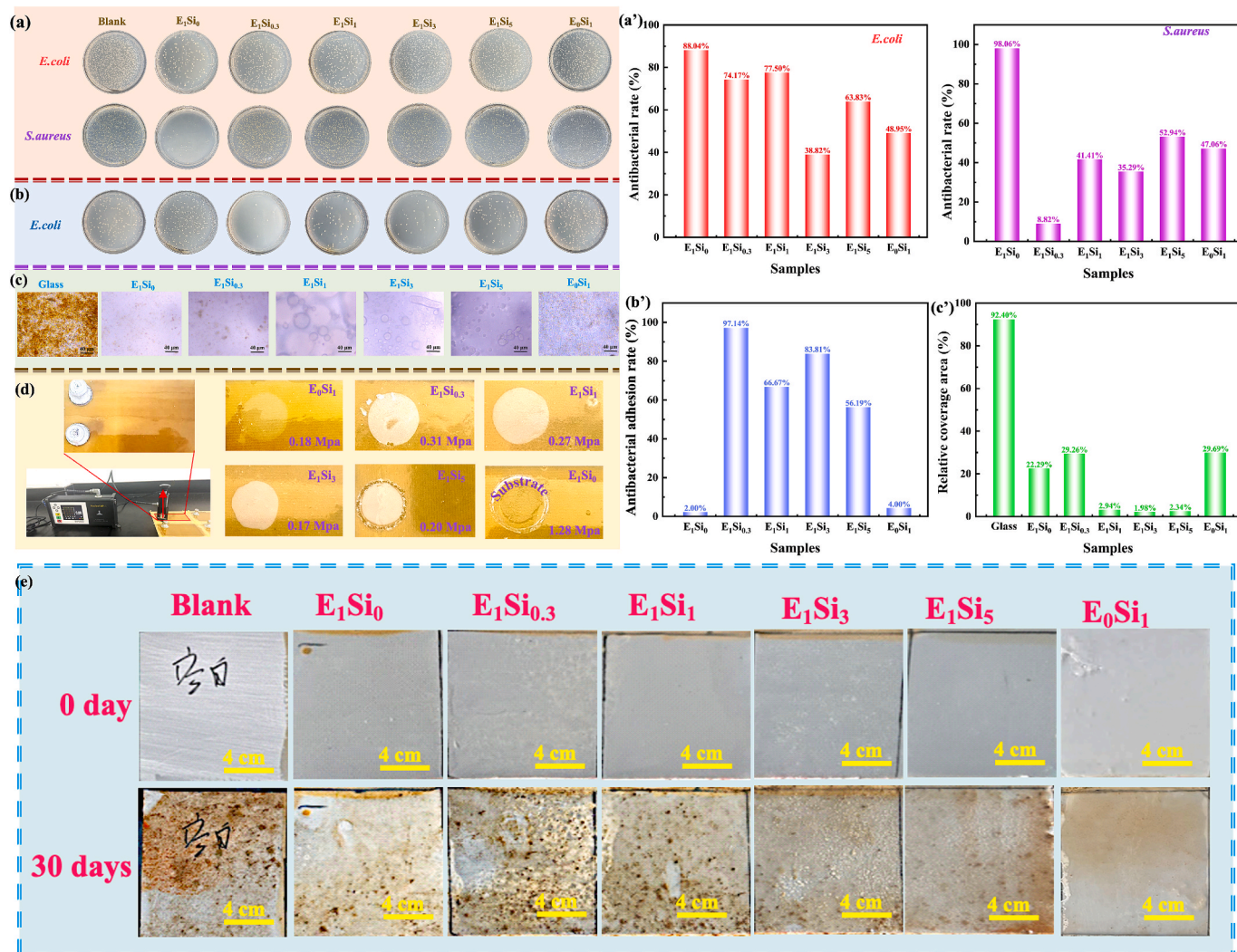


Fig. 8. (a) Images of the inhibitory effects of the E_xSi_y coating on *E. coli* and *S. aureus*. (a') The inhibition rate of the E_xSi_y coating on *E. coli*. (b) Images of the anti-adhesion effect of the E_xSi_y coating on *E. coli*. (b') The inhibition rate of the E_xSi_y coating on *S. aureus*. (c) Fluorescence microscope images of the E_xSi_y coating after immersion in the *N. closterium*. (c') The attachment area of *N. closterium* on the E_xSi_y coatings. (d) Images of the removal effect of Pseudobarnacle attachments on the E_xSi_y coating surface. (e) Marine antifouling performance test of E_xSi_y coatings in Qingdao Yellow Sea area (2024.12.19–2025.1.19).

4.7. Antifouling performance testing

In the initial stage of fouling organism growth, the surface of the coating is extremely prone to becoming a breeding ground for a large number of bacteria. To deeply explore the antibacterial ability of the E_xSi_y coating, we selected *E. coli* and *S. aureus* as typical microbial fouling organisms. As shown in Fig. 8(a) and 8(a'), due to the antibacterial properties of the EAM-Zn coating against *E. coli* and *S. aureus*, the E_xSi_y coating after compounding also has good antibacterial effects on these two kinds of bacteria. However, the introduction of silicone has significantly hindered the hydrolysis process of the coating, resulting in a substantial reduction in the original good antibacterial effect.

To further evaluate the comprehensive performance of the E_xSi_y coating, *E. coli* was used as the test bacterium, and the adhesion of it on the coating surface was tested. As illustrated by the results in Fig. 8(b) and 8(b'), the composite coating, owing to its outstanding self-cleaning property and hydrophobic surface, powerfully suppresses the adhesion of *E. coli* to the coating surface, achieving an antibacterial adhesion rate as high as 97.14 %.

During the early phase of fouling organism growth, in addition to the adhesion of bacteria, there exists a process of adhesion and growth of algal organisms. Fig. 8(c) and (c') demonstrate the attachment of

diatoms (*N. closterium*) on the coating surface, clearly presented through fluorescence microscope images and their corresponding coverage areas. Although diatoms tend to adhere to hydrophobic surfaces, the incorporation of EAM-Zn resin in E_xSi_y enables the release of zinc ions and eugenol upon hydrolysis in seawater, which significantly inhibit the growth of diatoms, thereby making it difficult for them to adhere to hydrophobic surfaces.

Furthermore, the smoothness of the coating surface is also a crucial factor influencing the adhesion strength of diatoms. Specifically, the roughness of $E_1Si_{0.3}$ and E_0Si_1 coatings is 5.82 μm and 3.82 μm respectively, making their surfaces relatively rough. Such surface conditions are conducive to diatom adhesion. Consequently, the coverage areas of diatoms on these two coatings are relatively large, reaching 29.26 % and 29.69 % respectively.

In the later stage of fouling organism growth, large organisms like barnacles tend to attach to the coating. Therefore, the fouling release performance of the coating is crucial as it determines the attachment of these large organisms. Fouling release performance was investigated through pseudobarnacle removal tests (Fig. 8d), where lower removal strength indicates superior fouling release ability. The removal strength of E_0Si_1 , $E_1Si_{0.3}$, E_1Si_1 , E_1Si_3 , E_1Si_5 and E_1Si_0 was approximately ~ 0.18 , ~ 0.31 , ~ 0.27 , ~ 0.17 , ~ 0.20 MPa and ~ 1.28 MPa, respectively,

indicating excellent fouling release ability attributable to the synergistic effect of lower SFE and surface roughness[49,50].

In summary, the E_xSi_y coating exhibits excellent antibacterial effects, outstanding antibacterial adhesion, strong anti-diatom performance, and good fouling release performance. Given these excellent properties, we carried out a marine field test by placing the E_xSi_y coating in the Yellow Sea area of Qingdao, as illustrated in Fig. 8(e). The results show that the coating has stably maintained antifouling effects for over a month in the actual marine environment, demonstrating its great potential for application in marine antifouling.

5. Conclusions

To effectively improve the inhibitory effect of the organosilicon antifouling coating on contaminants, this paper skillfully employs SIPNs to combine the EAM-Zn with biocidal effects and organosilicon. Meanwhile, the introduction of the highly elastic and strongly hydrophobic organosilicon material can effectively decelerate the hydrolysis rates of eugenol, zinc ions, and silicone esters, thereby reducing harm to the marine ecological environment.

Experimental result shows that the water absorption rate of the composite coating has dropped to 2 %, greatly reducing the hydrolysis rate of the coating in the real-sea environment and effectively enhancing the durability of the material. Additionally, the surface energy of the EAM-Zn coating has been successfully reduced to about 20 mJ/m², achieving the self-cleaning function of the coating and effectively reducing the possibility of marine organism attachment. Under the condition of EAM-Zn: Si = 1: 3, the composite coating has good flexibility, film-forming property, and smoothness ($R_a = 0.28 \mu\text{m}$), which can effectively solve the protection problem of the coating on irregular anti-fouling devices.

In terms of anti-fouling performance, the inhibition rate of the composite E_1Si_1 coating against *E. coli* has significantly increased from 48.95 % to 77.50 %. The surface energy of the E_xSi_y coating remains at 20 mJ/m², the adhesion area of diatoms on the surface of this coating is only 2 %, and the removal strength of pseudobarnacle is 0.2 MPa. Notably, it also demonstrated outstanding anti – fouling performance during actual sea area tests.

In conclusion, the coating formed by compounding organosilicon materials with EAM-Zn through SIPNs integrates the advantages of low surface energy, favorable hydrolysis performance, and high flexibility. This innovative approach offers a novel method for the development of multifunctional new-type antifouling coatings, effectively alleviating the irreversible harm that self-polishing antifouling coatings cause to the marine environment.

CRediT authorship contribution statement

Zheng Wang: Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Conceptualization. **Shuhong Nie:** Writing – review & editing, Supervision, Data curation. **Enhui He:** Writing – review & editing, Methodology, Investigation, Formal analysis. **Guanglei Song:** Visualization, Validation. **Xiaohui Jiang:** Writing – review & editing, Project administration. **Xia Li:** Writing – review & editing, Supervision, Resources. **Xuefeng Yan:** Supervision, Resources, Project administration. **Liangmin Yu:** Supervision, Resources, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2025.163181>.

Data availability

Data will be made available on request.

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